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Harmonic Balance Domain Decomposition Finite Elements for Nonlinear Passive Microwave Devices Analysis

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Abstract *Full-wave analysis of the time-harmonic field originated in waveguide devices containing nonlinear materials, possibly anisotropic, is addressed in this article. The approach combines finite elements and harmonic balance handling the nonlinearity via either Picard or relaxed iterations. To speed-up computation, domain decomposition is exploited to restrict the iteration to a small subdomain.*

Keywords domain decomposition method, finite-element methods, harmonic analysis, nonlinear media, waveguide filters

1. Introduction

Full-wave electromagnetic analysis of passive linear microwave devices can be performed via several well-established numerical methods. More complex devices, for example, those containing nonlinear, possibly anisotropic, materials, on the other hand, are a matter of active research. Within the broad subdivision among time-domain and frequency-domain methods, the former can handle nonlinearities straightforwardly, while the second require a more complex multi-harmonic formulation. Yet, it has been proven that time-domain analysis of waveguide devices, including nonlinear materials, often results to be computationally prohibitive (De Gersem et al., 2001).

As already mentioned, a steady-state frequency-domain simulation requires taking into account several harmonics at the same time to model all relevant phenomena. If the frequency-domain finite-element (FE) method is used, then a harmonic balance (HB) FE technique (Yamada et al., 1989) or multi-harmonic FE method (Copeland & Langer, 2010) becomes mandatory. In both references, low-frequency fields are considered, and in the latter, a substructuring domain decomposition (DD) technique was also exploited; DD is a useful technique allowing subdivision of the whole domain of the problem into smaller sub-domains, each of which is coupled to the others by appropriate boundary conditions. Each domain can then be solved separately with a reduced memory and CPU time requirement within some iterative framework up to obtain the complete solution (Quarteroni & Valli, 1999; Toselli & Widlund, 2004).

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Applications to higher frequencies is more recent and has been done first without resorting to HB (Guarnieri et al., 2010) and then by adding HB, but the analysis was limited to nonlinear, isotropic permittivities (Ntibarikure et al., 2012).

In this article, materials that can be nonlinear in both permittivity and permeability will be addressed, as well as anisotropic nonlinear materials. Also, the solution of the systems deriving from the DD will be handled with more efficient solvers. The capabilities of the code will be shown on a few H -plane device geometries.

The article is organized as follows. Section 2 presents the HB-FE formulation, while Section 3 details the DD technique. Section 4 presents some numerical results, and Section 5 draws conclusions.

2. Formulation

Consider a generic multiport device; in general, such a device must be analyzed via a full 3D numerical technique, but, in many practical cases, when there is no variation of the geometry in one direction, the analysis domain can be reduced to 2D (Figure 1).

The device may contain nonlinear anisotropic dielectric characterized by

$$\begin{aligned}\mathbf{D} &= [\varepsilon_0 \bar{\bar{\varepsilon}}_r(\mathbf{E}, \mathbf{H})] \mathbf{E}, \\ \mathbf{B} &= [\mu_0 \bar{\bar{\mu}}_r(\mathbf{E}, \mathbf{H})] \mathbf{H},\end{aligned}\tag{1}$$

with ε_0 and μ_0 being the permittivity and permeability of free space and $\bar{\bar{\varepsilon}}_r$ and $\bar{\bar{\mu}}_r$ the relative permittivity and permeability of the anisotropic, inhomogeneous medium contained in the device.

The field within the device is obedient to the vector Helmholtz equation:

$$\nabla \times (\bar{\bar{\mu}}_r^{-1}(\mathbf{E}(\mathbf{r}), \mathbf{r}) \nabla \times \mathbf{E}(\mathbf{r})) - k_0^2 \bar{\bar{\varepsilon}}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) = 0,\tag{2}$$

where on domain Ω , with perfect electric conductor boundary conditions on the device walls Γ_0 and a suitable modal expansion on the device ports Γ_l ($l = 1, \dots, L$), L is the number of ports (Pelosi et al., 2009).

The FE solution of Eq. (2) is well known and is obtained either by a variational formulation or a Galerkin procedure by expanding the unknown field $\mathbf{E}(\mathbf{r})$ onto a set of

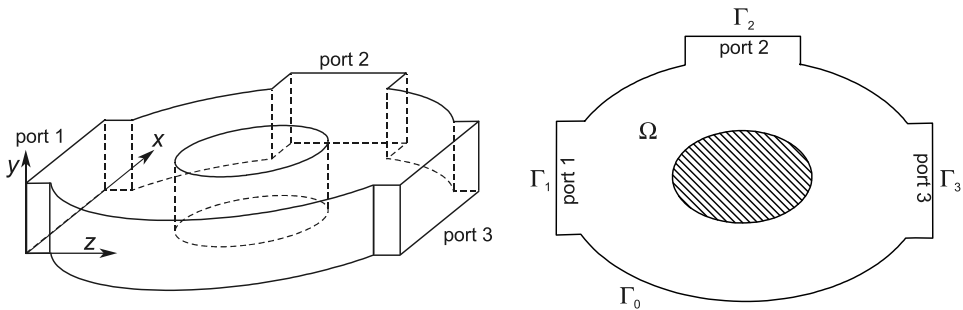


Figure 1. Generic multi-port device (left) with geometry uniform along z and corresponding 2D domain (right) on which full-wave analysis can be performed.

basis functions defined over a suitable discretization of the domain Ω in elements:

$$\mathbf{E}(\mathbf{r}) \cong \sum_{n=1}^N E_n \boldsymbol{\alpha}_n(\mathbf{r}). \quad (3)$$

In any case, a linear system where the unknowns are the coefficients $[E_1, E_2, \dots, E_N]$ of the expansion is obtained (Pelosi et al., 2009):

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ A_{21} & A_{22} & & A_{2N} \\ \vdots & & \ddots & \vdots \\ A_{N1} & A_{N2} & \cdots & A_{NN} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_N \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_N \end{bmatrix}. \quad (4)$$

Equation (2) is valid for harmonic field at a fixed frequency f_0 , and hence for a given value of $k_0 = 2\pi f_0 \sqrt{\epsilon_0 \mu_0}$, if the medium is nonlinear and higher-order harmonics are originated by it; hence, there is a set of equations like Eq. (2) to be solved simultaneously, each for a different frequency, coupled by the medium.

The total electric field can be written as a Fourier series:

$$\mathbf{E}(\mathbf{r}, t) = \sum_{p=1}^{\infty} \left(\mathbf{E}_p^{(s)}(\mathbf{r}) \sin(\omega_p t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(\omega_p t) \right), \quad (5)$$

with $\omega_p = p\omega_0 = 2p\pi f_0$, corresponding to $k_p = pk_0$.

If the incident field contains Q frequencies f_q ($q = 1, \dots, Q$), then the expansion of the electric field is

$$\mathbf{E}(\mathbf{r}, t) = \sum_{\{\mathbf{p}\}} \left(\mathbf{E}_{\{\mathbf{p}\}}^{(s)}(\mathbf{r}) \sin(\omega_{\{\mathbf{p}\}} t) + \mathbf{E}_{\{\mathbf{p}\}}^{(c)}(\mathbf{r}) \cos(\omega_{\{\mathbf{p}\}} t) \right), \quad (6)$$

which has essentially the same form as Eq. (3) except that the summation is extended to all possible Q -tuples $\{\mathbf{p}\} = \{p_1, p_2, \dots, p_Q\}$ with $p_q \in \mathbb{N}$ and $\omega_{\{\mathbf{p}\}} = p_q \omega_q$ with $\omega_q = 2\pi f_q$.

Since material parameters can also be expressed in terms of a Fourier series holding both for the case in Eq. (4) and for the case in Eq. (3), if $Q = 1$ and the Q -tuple reverts to a single integer (Yamada et al., 1989; Copeland & Langer, 2010),

$$[\bar{\epsilon}_r(\mathbf{E}, \mathbf{H}), \bar{\mu}_r(\mathbf{E}, \mathbf{H})] = \sum_{\{\mathbf{g}\}} \left([\bar{\epsilon}_{r_{\{\mathbf{g}\}}}^{(s)}, \bar{\mu}_{r_{\{\mathbf{g}\}}}^{(s)}] \sin(\omega_{\{\mathbf{g}\}} t) + [\bar{\epsilon}_{r_{\{\mathbf{g}\}}}^{(c)}, \bar{\mu}_{r_{\{\mathbf{g}\}}}^{(c)}] \cos(\omega_{\{\mathbf{g}\}} t) \right), \quad (7)$$

with

$$\begin{aligned} [\bar{\epsilon}_{r_{\{\mathbf{g}\}}}^{(s)}, \bar{\mu}_{r_{\{\mathbf{g}\}}}^{(s)}] &= \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} [\bar{\epsilon}_r(\mathbf{E}, \mathbf{H}), \bar{\mu}_r(\mathbf{E}, \mathbf{H})] \sin(\omega_{\{\mathbf{g}\}} t) dt, \\ [\bar{\epsilon}_{r_{\{\mathbf{g}\}}}^{(c)}, \bar{\mu}_{r_{\{\mathbf{g}\}}}^{(c)}] &= \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} [\bar{\epsilon}_r(\mathbf{E}, \mathbf{H}), \bar{\mu}_r(\mathbf{E}, \mathbf{H})] \cos(\omega_{\{\mathbf{g}\}} t) dt. \end{aligned} \quad (8)$$

The infinite summations in Eqs. (6) and (7) are truncated to a finite number of terms, respectively, P and G , to allow for numerical computation. A possible truncation criterion, as illustrated in Section 4, is to limit the harmonic expansions such that the impinging power at fundamental frequencies does not vary more than a prescribed value while the order is increased (Ntibarikure et al., 2012). In this framework, Eq. (2) must be solved for all harmonic fields $\mathbf{E}_{\{\mathbf{p}\}}^{(s,c)}$ in the expansion in Eq. (6); hence, the expansion in Eq. (3) must be applied to all $\mathbf{E}_{\{\mathbf{p}\}}^{(s,c)}$, and the linear system in Eq. (4) stays formally the same except that each entry A_{mn} of matrix $[\mathbf{A}]$ itself becomes a $2P \times 2P$ matrix, while each entry E_n of vector $[\mathbf{E}]$ becomes a $2P$ vector, relative to the P sine and P cosine coefficients in Eq. (6). The same holds true for the known vector $[\mathbf{B}]$ terms:

$$A_{mn} = \begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \cdots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \cdots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \cdots & a_{mn}^{(PcPc)} \end{bmatrix}, \quad E_n = \begin{bmatrix} e_n^{(1s)} \\ e_n^{(1c)} \\ \vdots \\ e_n^{(Ps)} \end{bmatrix}, \quad B_m = \begin{bmatrix} b_m^{(1s)} \\ b_m^{(1c)} \\ \vdots \\ b_m^{(Pc)} \end{bmatrix}, \quad (9)$$

where indices p and q in $a_{mn}^{(p[s,c]q[s,c])}$ are relative to an appropriate ordered indexing of all the possible Q -tuples $\{\mathbf{p}\}$. In the case of a single incident frequency, then p is the order of the harmonic.

Since $a_{mn}^{(p[s,c]q[s,c])}$, and hence A_{mn} , depends on the material properties in Eqs. (1) and (2), and hence on the electric and magnetic fields, it is evident that entries of matrix $[\mathbf{A}]$ depend on the solution $[\mathbf{E}]$, and thus, a direct solution is not possible.

An iterative solution is hence necessary, hypothesizing a null first guess $[\mathbf{E}]^{(0)}$ for the unknown electric field to compute a first system matrix $[\mathbf{A}([\mathbf{E}]^{(0)})]$ and right-hand side $[\mathbf{B}([\mathbf{E}]^{(0)})]$. From these, a new value of the unknown vector is obtained by solving $[\mathbf{A}([\mathbf{E}]^{(0)})][\mathbf{E}]^{(1)} = [\mathbf{B}([\mathbf{E}]^{(0)})]$. From the new value $[\mathbf{E}]^{(1)}$, a new system can be computed.

Rather than the simple Picard iteration detailed above, a relaxed iteration often proves to have better convergence properties. In the relaxed iteration, the field used to compute the new linear system is a weighted sum of the two previous iteration step results:

$$[\mathbf{A}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}][\mathbf{E}]^{(i)} = [\mathbf{B}_{(\gamma[\mathbf{E}]^{(i-1)} + (1-\gamma)[\mathbf{E}]^{(i-2)})}], \quad i \geq 2, \quad (10)$$

with $\gamma \in (0, 1]$. $\gamma = 1$ gives the standard Picard iteration (Guarnieri et al., 2010), while low values damp the oscillations that may arise with highly nonlinear materials or high intensity impressed fields at the cost of a slower convergence (Silvester & Ferrari, 1997). The process is repeated until the relative error between the updated solution and the previous one, in the sense of the Euclidean norm, is less than a prescribed value (Guarnieri et al., 2010; Ntibarikure et al., 2012):

$$\frac{\|[\mathbf{E}]^{(i)} - [\mathbf{E}]^{(i-1)}\|}{\|[\mathbf{E}]^{(i)}\|} \leq \tau. \quad (11)$$

The order of the resulting HB-FE systems in Eq. (10) is hence $2PN$, and this number can become very high, especially if more than a frequency is present in the impinging signal and intermodulation products (IMPs) are generated. Furthermore, since solutions have to

be iterated, building a new linearized system at each step, computing time can become too high. To enhance the computational efficiency, a DD approach can be exploited, as detailed in the following section.

3. Domain Decomposition

In the proposed DD approach the original problem domain Ω (Figure 1) is decomposed in several smaller subdomains Ω_i so that (Figure 2)

$$\Omega = \bigcup_{i=1}^N \Omega_i, \quad \Omega_i \cap \Omega_j = \emptyset, \quad \forall i \neq j. \quad (12)$$

The unknowns of the original FE problem (Eq. (4)) can be renumbered on a subdomain-by-subdomain basis to group together all unknowns internal to each subdomain while keeping together all the unknowns relative to all boundaries $\Gamma_0, \Gamma_l, \Gamma_{i,j}$.

By naming $[\mathbf{E}]_i$, the vector of the unknowns internal to Ω_i , $[\mathbf{E}]_d = [[\mathbf{E}]_1, [\mathbf{E}]_2, \dots, [\mathbf{E}]_N]$ the vector of all unknowns internal to all subdomain, and $[\mathbf{E}]_c$ the vector of the unknowns on all boundaries, and by performing a similar operation on the right-hand side term, Eq. (4) can be rewritten as

$$[\mathbf{A}][\mathbf{E}] = [\mathbf{B}] \Leftrightarrow \begin{bmatrix} [\mathbf{M}] & [\mathbf{P}] \\ [\mathbf{Q}] & [\mathbf{C}] \end{bmatrix} \begin{bmatrix} [\mathbf{E}]_d \\ [\mathbf{E}]_c \end{bmatrix} = \begin{bmatrix} [\mathbf{B}]_d \\ [\mathbf{B}]_c \end{bmatrix}$$

$$\Leftrightarrow \left[\begin{array}{cccc|c} [\mathbf{M}]_1 & 0 & \dots & 0 & [\mathbf{P}]_1 \\ 0 & [\mathbf{M}]_2 & \dots & 0 & [\mathbf{P}]_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & [\mathbf{M}]_N & [\mathbf{P}]_N \\ \hline [\mathbf{Q}]_1 & [\mathbf{Q}]_2 & \dots & [\mathbf{Q}]_n & [\mathbf{C}] \end{array} \right] \begin{bmatrix} [\mathbf{P}]_1 \\ [\mathbf{P}]_2 \\ \vdots \\ [\mathbf{E}]_N \\ [\mathbf{E}]_c \end{bmatrix} = \begin{bmatrix} [\mathbf{B}]_1 \\ [\mathbf{B}]_2 \\ \vdots \\ [\mathbf{B}]_N \\ [\mathbf{B}]_c \end{bmatrix}, \quad (13)$$

where the block structure of matrix $[\mathbf{A}]$ is evident. The form in Eq. (13) of course holds also for the multi-harmonic case if each entry in $[\mathbf{C}]$ is itself a matrix (Eq. (9)).

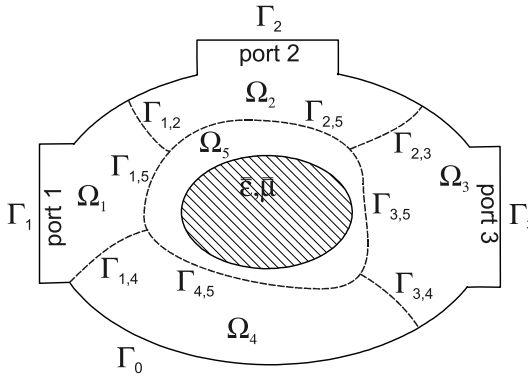


Figure 2. Generic multi-port device with geometry uniform along z and a possible subdivision in five subdomains.

If the domains are arranged so that the nonlinear anisotropic material is fully contained in the last domain Ω_N , then Eq. (13) can be rewritten with its explicit dependence on the iteration (Eq. (10)) as

$$\begin{aligned}
 & \begin{bmatrix} [\mathbf{M}]_1 & 0 & \cdots & 0 & [\mathbf{P}]_1 \\ 0 & [\mathbf{M}]_2 & \vdots & \vdots & [\mathbf{P}]_2 \\ \vdots & & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & [\mathbf{M}_{(\gamma[\mathbf{E}]^{i-1}+(1-\gamma)[\mathbf{E}]^{i-2})N} & [\mathbf{P}_{(\gamma[\mathbf{E}]^{i-1}+(1-\gamma)[\mathbf{E}]^{i-2})N} \\ [\mathbf{Q}]_1 & [\mathbf{Q}]_2 & \cdots & [\mathbf{Q}_{(\gamma[\mathbf{E}]^{i-1}+(1-\gamma)[\mathbf{E}]^{i-2})N} & [\mathbf{C}_{(\gamma[\mathbf{E}]^{i-1}+(1-\gamma)[\mathbf{E}]^{i-2})} \end{bmatrix} \begin{bmatrix} [\mathbf{E}]_1^{(i)} \\ [\mathbf{E}]_2^{(i)} \\ \vdots \\ [\mathbf{E}]_N^{(i)} \\ [\mathbf{E}]_c^{(i)} \end{bmatrix} \\
 &= \begin{bmatrix} [\mathbf{B}]_1 \\ [\mathbf{B}]_2 \\ \vdots \\ [\mathbf{B}_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})N} \\ [\mathbf{B}_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})c} \end{bmatrix}. \quad (14)
 \end{aligned}$$

In Eq. (14), it is evident how the larger part of $[\mathbf{A}]$ does not change during the iteration, with updating being limited to the four blocks in the lower right corner.

By using the Schur complement concept (Saad, 2000), the boundary unknowns of Eq. (14), and hence the generalized scattering matrix of the device, can be retrieved with the same accuracy of the full domain solution (Guarnieri et al., 2010; Ntibarikure et al., 2012):

$$[\mathbf{S}]^{(i-1)}[\mathbf{E}]_c^{(i)} = [\mathbf{D}]^{(i-1)}, \quad (15)$$

with

$$\begin{aligned}
 [\mathbf{S}]^{(i-1)} &= [\mathbf{C}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})}] - \left\{ \sum_{n=1}^{N-1} [\mathbf{Q}]_n [\mathbf{M}]_n^{-1} [\mathbf{P}]_n \right\} \\
 &\quad - [\mathbf{Q}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})N}] [\mathbf{M}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})N}]^{-1} \\
 &\quad \cdot [\mathbf{P}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})N}] \quad (16)
 \end{aligned}$$

and

$$\begin{aligned}
 [\mathbf{D}]^{(i-1)} &= [\mathbf{B}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})}] - \left\{ \sum_{n=1}^{N-1} [\mathbf{Q}]_n [\mathbf{M}]_n^{-1} [\mathbf{P}]_n \right\} \\
 &\quad - [\mathbf{Q}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})N}] [\mathbf{M}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})N}]^{-1} \\
 &\quad \cdot [\mathbf{B}_{(\gamma[E]^{(i-1)}+(1-\gamma)[E]^{(i-2)})N}]. \quad (17)
 \end{aligned}$$

In Eqs. (16) and (17) the terms in braces, containing computationally intensive inversions, need to be computed only once before starting the iteration. Thus, to solve the HB-DD-FE system, every single submatrix is assembled at first sight; then, within the iteration loop, only the submatrices related to Ω_N and Γ , and hence the first and last term in the right-hand side of Eqs. (16) and (17), are to be updated.

The efficiency of this DD technique relies on the restriction of matrices updating to sufficiently small regions along the nonlinear iterative solution, noticeably improving long nonlinear loops. The Schur complement matrices related to linear materials domains, remaining constant all over the iterative solution loop, have to be assembled only once. The computation of the Schur complements with conventional techniques leads to a small but dense matrix (Toselli & Widlund, 2004; Guarnieri et al., 2010), which is known for leading to high computational complexity for large nonlinear subdomains. One way to reduce the complexity of the Schur complement-based DD could be the use of an iterative solver, with a proper preconditioner, for the boundary unknowns solution (Gaidamour & Henon, 2008).

4. Numerical Results

To assess the validity of the proposed approach, the test case of an H -plane circulator in a rectangular waveguide constituted by a Y-junction with a magnetized ferrite post is presented (Koshiba & Suzuki, 1986).

The circulator comprises three WR90 (cross-section dimensions: $a = 22.86$ mm, $b = 10.16$ mm) waveguide sections of length $l = 33.4$ mm, joined by forming a 120° angle one with each other in a Y shape, and a magnetized ferrite post of height b , hence spanning the whole device height. The post is an equilateral triangular prism of base side $d = 7.80$ mm, placed at the center of the junction as sketched in Figure 3, and it is magnetized along z .

While most of the device, Ω_1 , is in air ($\varepsilon_r = \mu_r = 1$), the post Ω_2 is made of a nonlinear magnetized ferrite. In Koshiba and Suzuki (1986) and many other papers, the

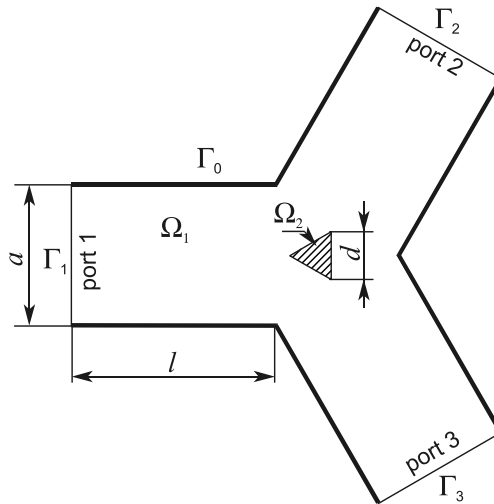


Figure 3. Geometry layout of the ferrite circulator. The two domains Ω_1 and Ω_2 for the DD approach are also shown.

ferrite is considered linear and characterized by

$$\bar{\bar{\mu}}_r(\mathbf{r}) = \begin{bmatrix} \mu_r & j\kappa_r & 0 \\ -j\kappa_r & \mu_r & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \bar{\bar{\epsilon}}_r(\mathbf{r}) = \epsilon_r, \quad (18)$$

with, as in the standard linear approach in which only the static (DC) external impressed magnetic field is considered and the harmonic (AC) signal neglected,

$$\mu_r = 1 + \frac{(\omega_0 + j\omega\alpha)\omega_m}{(\omega_0 + j\omega\alpha)^2 - \omega^2},$$

$$\kappa_r = \frac{\omega\omega_m}{(\omega_0 + j\omega\alpha)^2 - \omega^2}, \quad (19)$$

with $\omega_0 = \gamma H_i$, $\omega_m = \gamma M_s$, $\alpha = \gamma \Delta H / 2\omega$, and H_i being the external DC magnetic impressed field (uniform along the out of plane direction), M_s the saturation magnetization, ΔH the resonance linewidth, and γ the gyromagnetic ratio. In the following, the values considered will be $\gamma = 1.76 \cdot 10^7$ Ckg⁻¹, $H_i = 200$ G, $M_s = 1,317$ Oe, $\Delta H = 135$ Oe-s, and $\epsilon_r = 11.7$, as reported in Koshiba and Suzuki (1986).

Equation (18) still holds for nonlinear ferrites if the AC magnetic field is not neglected with respect to the external DC magnetic field. In this case, $\mathbf{H} = H_i \hat{z} + \mathbf{h}_{AC}$, and $\mathbf{M} = M_s \hat{z} + \mathbf{m}_{AC}$; upon derivation from the macroscopic equation of motion for large signals and assumption of the z -directed components of the AC field to be neglectable, Eqs. (19) are replaced by

$$\mu_r = 1 + \frac{(\omega_0 + j\omega\alpha)\omega_m}{(\omega_0 + j\omega\alpha)^2 + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2},$$

$$\kappa_r = \frac{\omega\omega_m}{(\omega_0 + j\omega\alpha)^2 + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2}, \quad (20)$$

where h_x and h_y are the two components on the H -plane of the AC magnetic field.

The analysis is first conducted assuming small signals impinging on Port 1 (Γ_1), hence with linear permeability tensor. The resulting spectral response is given in Figure 4, and it matches the one reported by Koshiba and Suzuki (1986). Figure 5 reports the corresponding fieldmap for a 9-GHz signal.

To assess nonlinearity of the modeled device, the circulator is analyzed, assuming a strong interfering signal impinging on the isolated port (Γ_3) at $f_i = 10$ GHz, 10 dB higher than the signal at Port 1, with signal frequency $f_s \in [8, 12]$ GHz, engendering undesired IMPs at the coupled port (Γ_2). The convergence of the harmonic orders (P) is analyzed upon increasing the order progressively and monitoring the power kept by in-band IMP signals ($2f_i - f_s$ and $2f_s - f_i$). The order of G is chosen such that $G\omega_g = \max(\omega_{\{p\}})$ and $\omega_g = \gcd(\omega_i, \omega_s)$. For this test, the signal power impinging at $f_s = 9$ GHz has been assumed to be 150 W, and the interferer is 1.5 kW at $f_i = 10$ GHz. An iteration loop

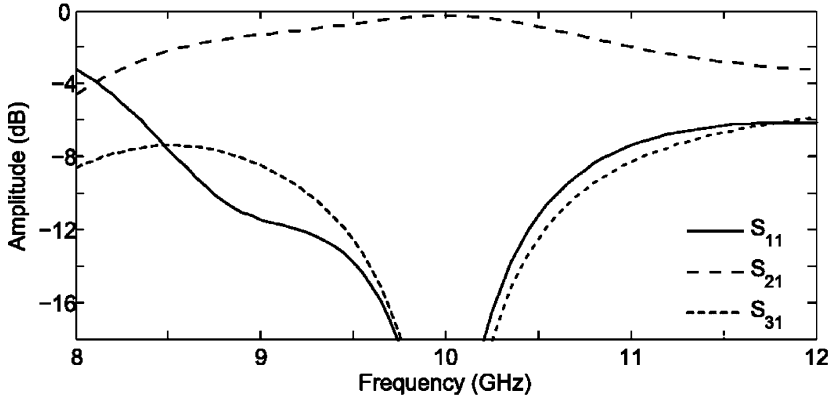


Figure 4. Circulator's small signal frequency response.

stop criterion $\tau = 10^{-9}$ has been assumed. Results are shown in Table 1, while Figure 6 reports the fieldmaps at the signal and interferent frequencies, as well as the in-band interferent.

It is worth noticing that as the harmonic order grows, higher-order TE modes at port boundaries become propagating, and hence, ten modes (Pelosi et al., 2009) have been retained to accurately compute the power balance.

Even if the exact (continuous) solution cannot be achieved, Table 1 shows that the computation of in-band third-order products, considering up to all third-order products, leads to a maximum error of less than 0.2%. In this case, Figures 7 and 8 show the

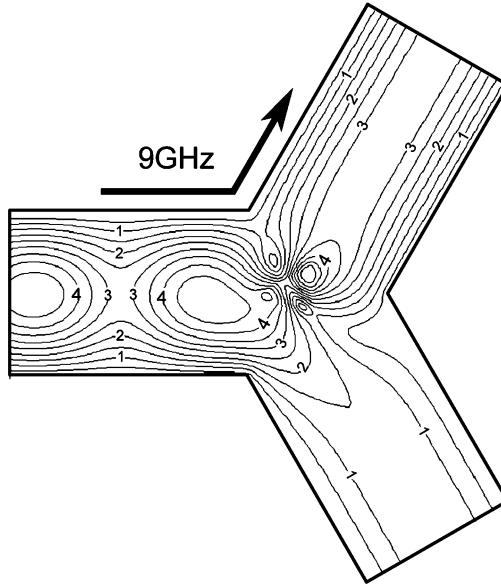


Figure 5. Electric field (kV/m) distribution at 9 GHz for 1 W impinging on Γ_1 .

Table 1
Harmonic power at third-order IMPs as P is increased

Harmonic order	Harmonic order considered (cumulative from left to right)			
	Only $2f_1 - f_2$, $2f_2 - f_1$	$2f_1 + f_2$, $2f_2 + f_1$, $3f_1, 3f_2$ (all third order)	$f_1 + f_2$, $ f_1 - f_2 $, $2f_1, 2f_2$ (all second order)	$3f_1 - 2f_2$, $3f_2 - 2f_1$, $3f_1 + 2f_2$, $3f_2 + 2f_1$
Power at $2f_1 - f_2$ (W)	4.6019×10^{-3}	4.2246×10^{-3}	4.2246×10^{-3}	4.2186×10^{-3}
Power at $2f_2 - f_1$ (W)	2.0990×10^{-1}	2.1951×10^{-1}	2.1951×10^{-1}	2.1991×10^{-1}

resulting power amplitudes as a function of frequency for the two in-band third-order IMPs for different signal and interferent power levels. The power delivered to IMPs is in good agreement with those reported in How et al. (1997).

The use of the DD scheme, which provides the same solution of the standard full system iteration scheme, allows speed up of computations. The acceleration of the DD solver, defined as the ratio between the standard scheme times (assembly and solve for whole domain at each iteration) and the DD times (assembly and solve for whole domain only at first iteration, then assembly and solve for nonlinear subdomains only) is reported in Figure 9. Sparse direct solvers of a MATLAB® (The MathWorks, Natick, Massachusetts, USA) implementation for both standard full and DD schemes have been employed to perform the computations.

All over the frequency sweep, the fast Fourier transform (FFT) based algorithm employed to compute nonlinear materials testing has required a noticeably different amount of times, bound to differences in the number of iterations required, leading to different values of acceleration for the chosen frequency points. The time step required to accurately compute the testing integrals strongly depend on the difference between signal and interferer frequencies. However, 45% to 60% average speed-ups have been noticed within the signal bandwidth.

Further tests are performed to assess the efficiency of the method, both for two and five subdomains (only one with nonlinear ferrite), while increasing the number of degrees of freedom (hence the non-zero entries of the system matrix $[A]$) by increasing the elements order from first to fourth (Figures 10 and 11).

The higher the number of linear subdomains is, the smaller the submatrices of the linear subdomains are, leading to improved acceleration. Furthermore, acceleration grows logarithmically with the polynomial order, that is, with the number of non-zero entries of the system matrix.

Increasing the number of subdomains generally leads to higher allocated memory. This is mainly due to an increase of geometrical and degrees of freedom reordering infor-

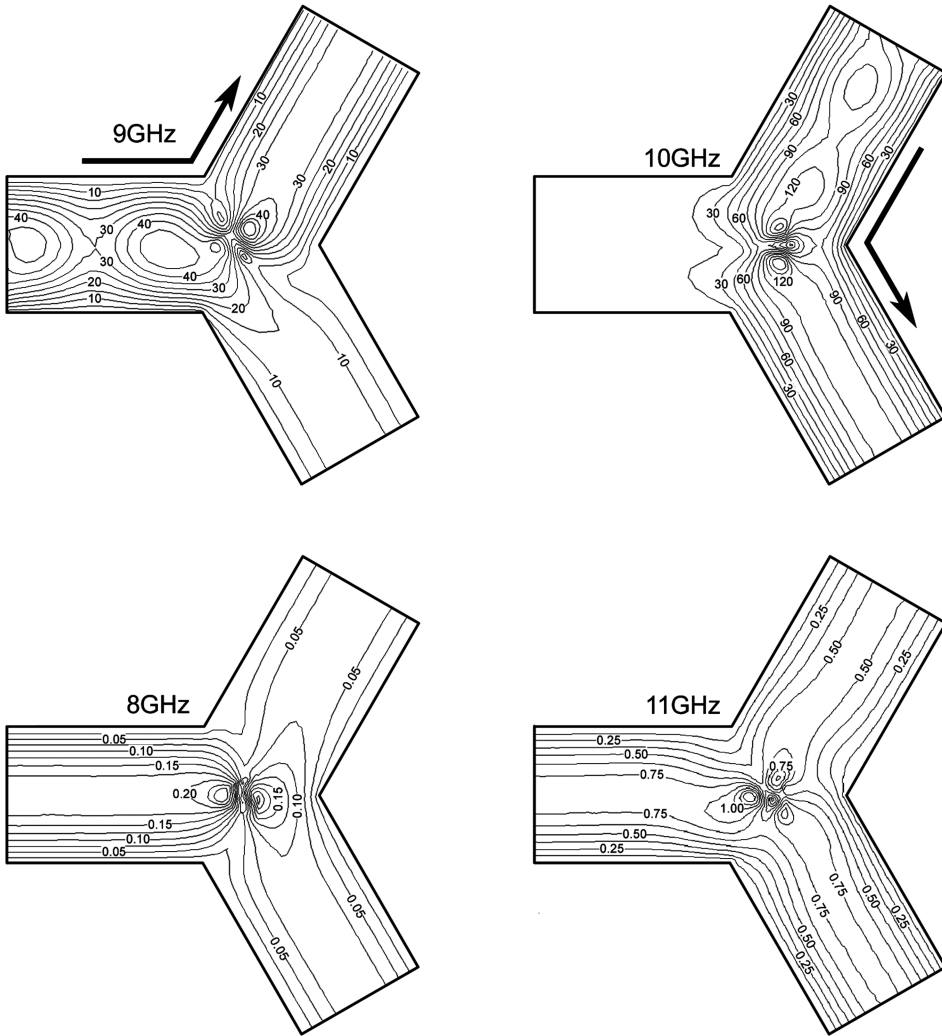


Figure 6. Electric field (kV/m) distribution for third-order intermodulation problem at 9 GHz, 10 GHz, and in-band IMPs frequencies 8 and 11 GHz.

mation, which is kept in both standard full and DD schemes. DD schemes require more memory to store the Schur complement matrices and solve for related unknowns.

5. Conclusions

An FE approach to the analysis of nonlinear passive microwave devices has been presented. The article presents HB as a valid methodology for the calculation of harmonic couplings occurring within nonlinear materials characteristics at microwave frequencies. A Schur complement-based DD technique has been applied to speed up the nonlinear iterative solution.

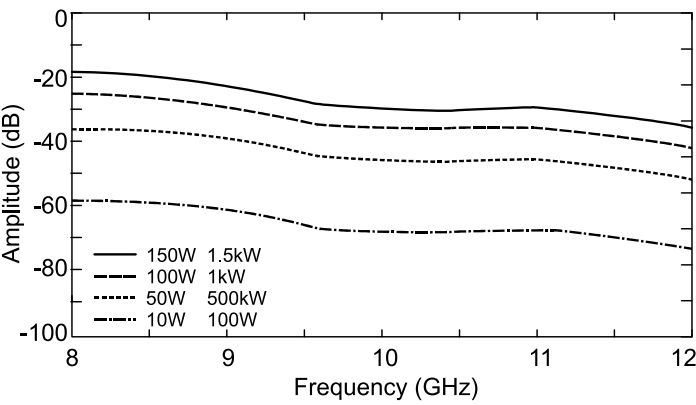


Figure 7. IMP power at $2f_s - f_i$.

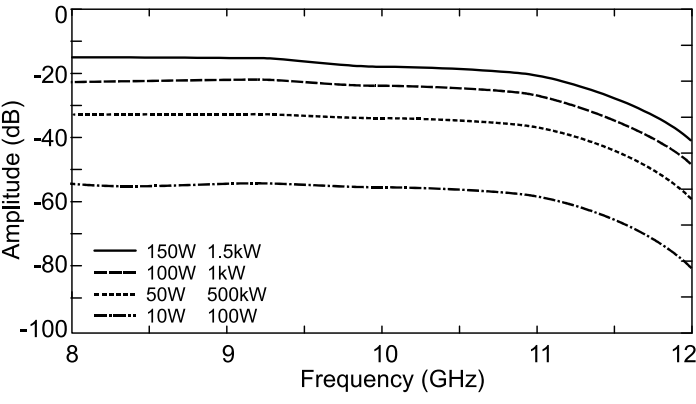


Figure 8. IMP power at $2f_i - f_s$.

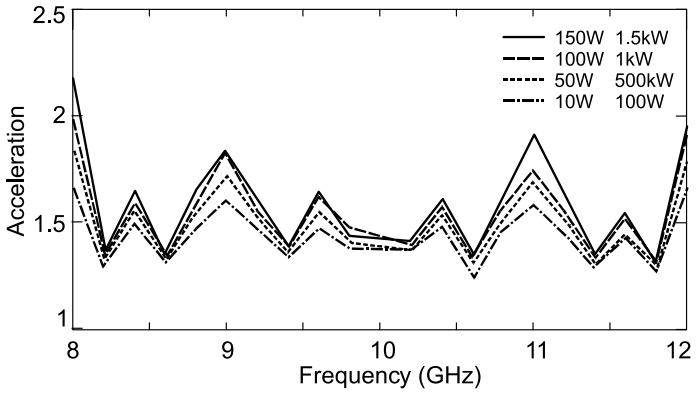


Figure 9. Acceleration of DD scheme versus conventional scheme. Linear elements have been considered, leading to 700 degrees of freedom and 8,100 non-zero entries of $[A]$.

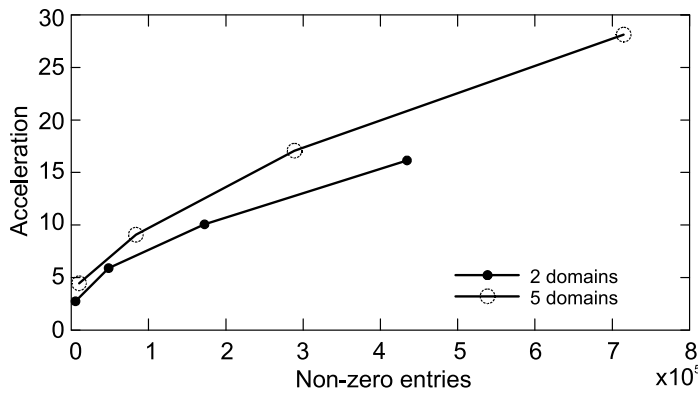


Figure 10. Acceleration of DD scheme versus standard full scheme as a function of nonzero entries of $[A]$.

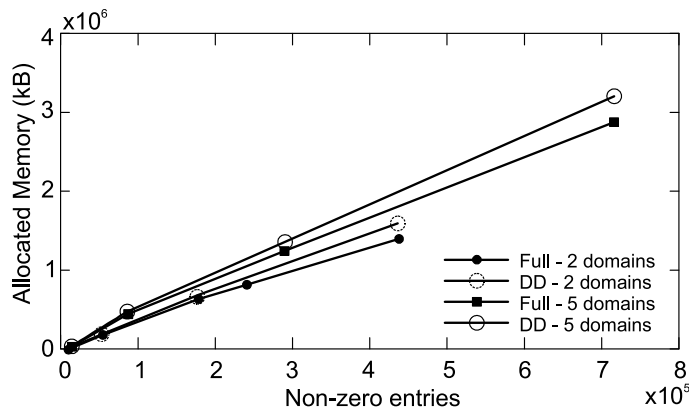


Figure 11. Allocated memory (stored and dynamically freed) as a function of non-zero entries of $[A]$.

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